Project Description: Simulated Distribution Station Control

Overview

This project involves developing and implementing an industrial automation control system for a simulated Distribution Station. The primary function of the station is to store raw workpieces and transfer them reliably to downstream production processes within a modular, distributed manufacturing environment.

System Architecture & Subsystems

The station is divided into two primary, interconnected subsystems that must execute coordinated control sequences:

1. Stack Magazine Subsystem

- **Function:** Stores workpieces in a vertical column and dispenses them one by one.
- **Controller Responsibilities:**
    - Empty sensor
    - Cylinder limit position sensors
    - Commands the cylinder to feed a workpiece

2. Transfer Unit Subsystem

- **Function:** Picks up the separated workpiece at the handover point and transports it to the defined output location for the next station.
- **Controller Responsibilities:**
    - Coordinates a vacuum gripper to grip and release parts safely.
    - Controls transfer drive actuators to transition between pickup and drop-off positions.
    - Motion limit switches
    - Vacuum sensor

3. Requirements

3.1 General Requirements

GR-01 Station should work in MANUAL and AUTOMATIC state.

3.2 Initial Requirements

IR-01 At Initial state Stack Magazine cylinder is retracted
IR-02 At Initial state Transfer Unit Arm is at magazine point and vacuum is turned off
IR-03 After system is turned on , station goes to initial position

3.3 Automatic Mode Requirements

Those requirements are suitable for automatic mode:

AMR-01 When Stack Magazine and Transfer Unit are in Initial position , Ready indicator should activate(Start button led turn on)
AMR-02 When Start button is pressed , station should start(continue) working cycle
AMR-03 When Stop is triggered, station should continue to next stable state and then stop working cycle
AMR-04 When station is stopped and RESET is triggered, station and acknowledge indication should go to initial state
AMR-05 If Magazine stack is not empty, Acknowledge button led should turn on
AMR-06 If Magazine stack is empty, Acknowledge button should start blink
AMR-07 Acknowledge button stops blinking, after operator pressed acknowledge button

3.4 Stack Magazine Requirements

MR-01 if sensor is empty for longer then 1.5 second, stack empty indicator should activate(ack blinking)

3.5 Transfer Unit Requirements

TR-01 Vacuum should not turn off during transportation of workpiece between stable points

3.6 Manual Mode Requirements

During manual operation next functionalities are available:
MMR-01: When Push WP is active, Stack Magazine cylinder is extending
MMR-02: When Push WP is not active, Stack Magazine cylinder is retracting
MMR-03 When GO TO MAG is active, arm goes to magazine point
MMR-04 When GO TO NEXT is active, arm goes to next station point
MMR-05 When both GO TO MAG and GO TO NEXT is active, arm stops
MMR-06 When Vacuum on is active, vacuum is active
MMR-07 When Vacuum off is active, vacuum is not active
MMR-08 When both VACUUM ON and VACUUM OFF is active, vacuum is not active

4.Logical Design

Stack magazine and Arm Transport Unit work simultaneously from each other that allows easy extending of the system with other processing units.

Stack Magazine

Stack magazine at the beginning enters Setup state and gives command to cylinder to retract. When it is retracted, it is set to initial position and indicates it is ready for Automatic workflow. Automatic cycle consist of 4 states : Extend, **Extended**, Retract, **Retracted**. Extended and Retracted are stable states where Stack Magazine can safely stop .

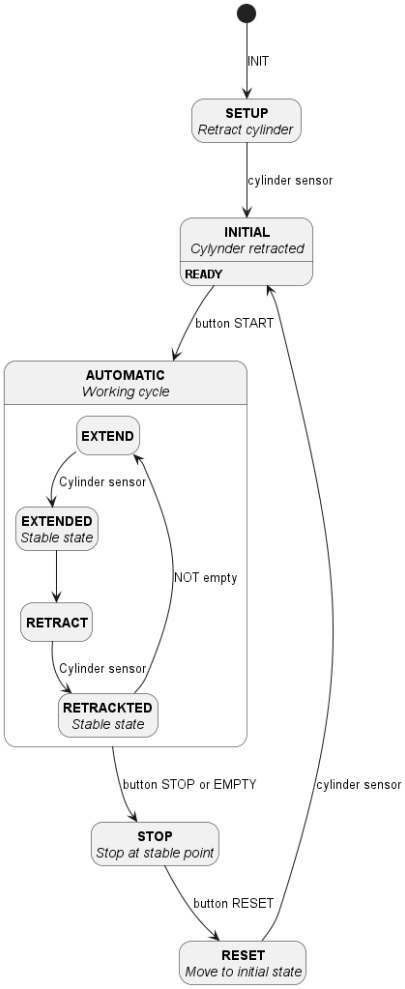


Figure 1Stack Magazine State diagram

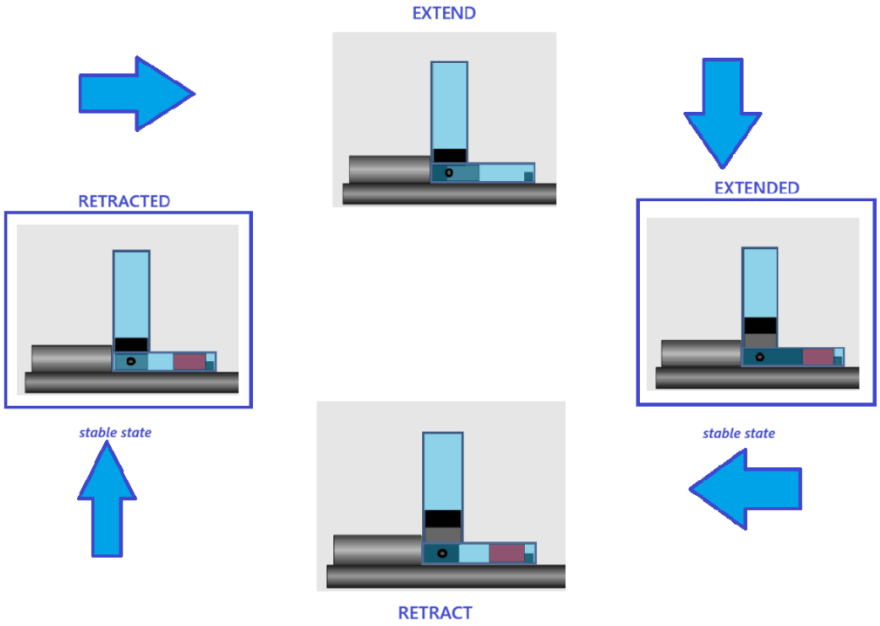


Figure 2 Stack Magazine automatic workflow

Arm Transfer Unit

Arm Transfer Unit at the beginning enters Setup state and gives arm command to go to next station point. In initial position controller indicates it is ready for Automatic workflow. Automatic cycle consist of 4 states : **Pick_up**, Picked_up, **Put_down**, To_magazine. Pick_up and Put_down are stable states where Arm can safely stop .

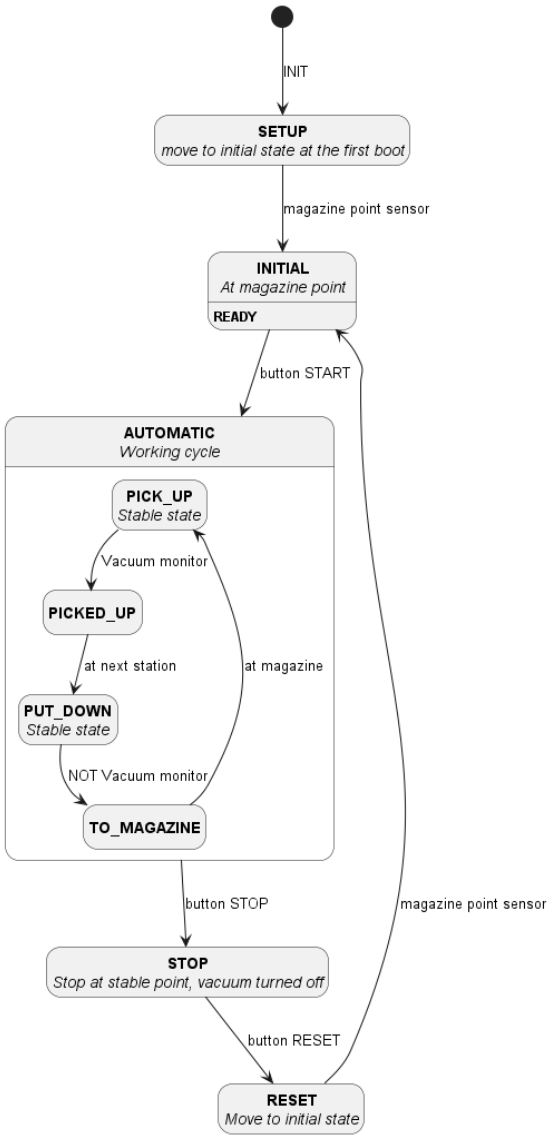


Figure 3 Arm Transport Unit State diagram

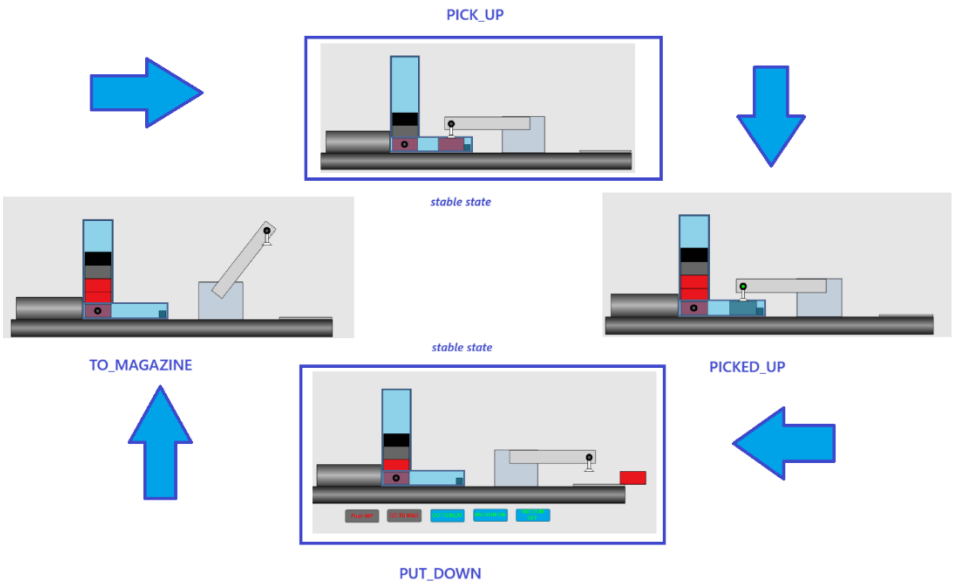


Figure 4 Arm Transport Unit automatic workflow

5. Implementation

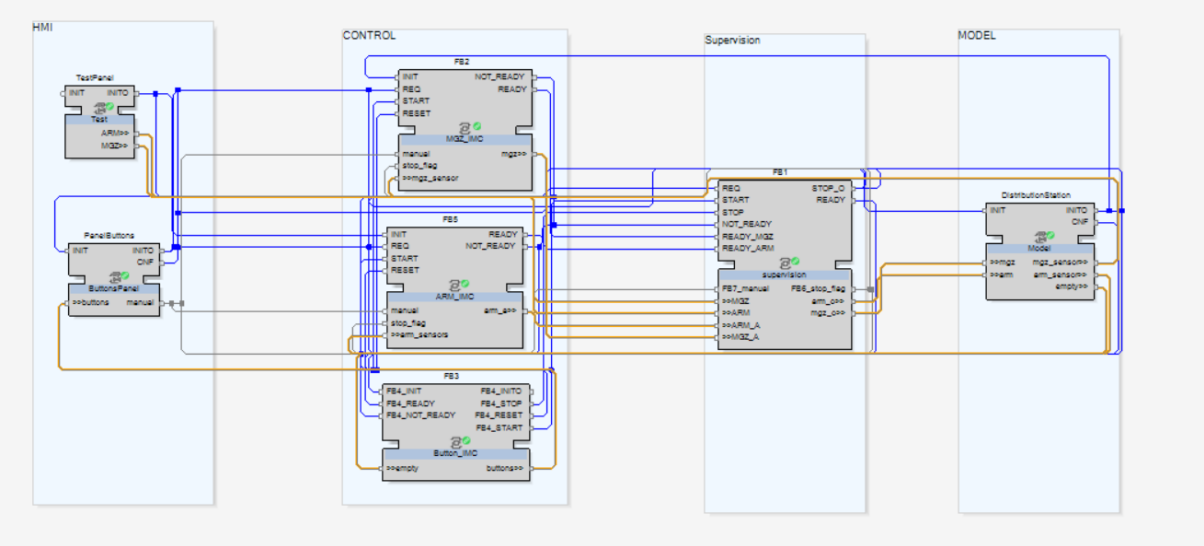


Figure 5 Application architecture

IMC pattern is used to implement software design. Application consists of HMI(Test CAT and ButtonsPannel CAT),Control (MGZ_IMC, ARM_IMC, Button_IMC), Supervision, Model.

ARM_IMC

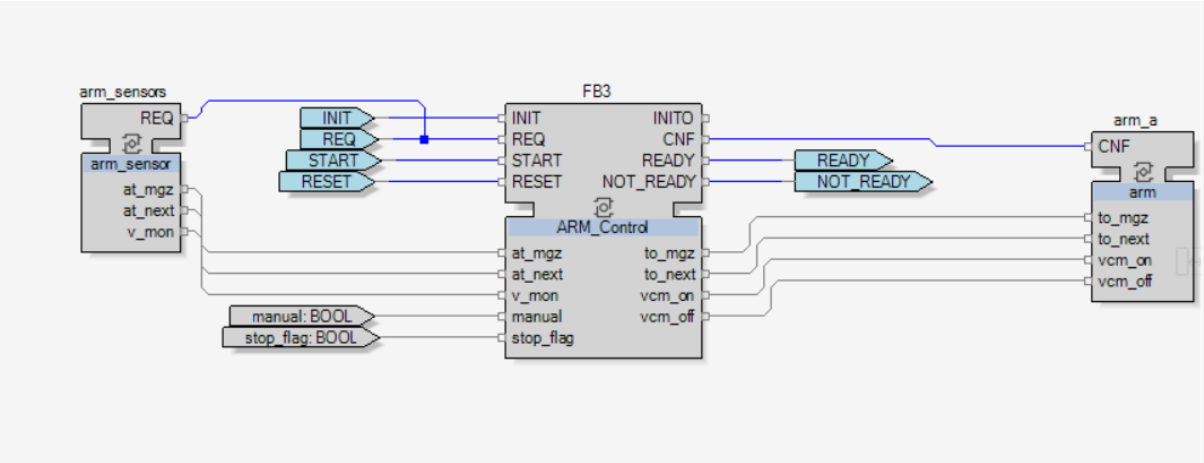


Figure 6 Arm Transport Unit IMC

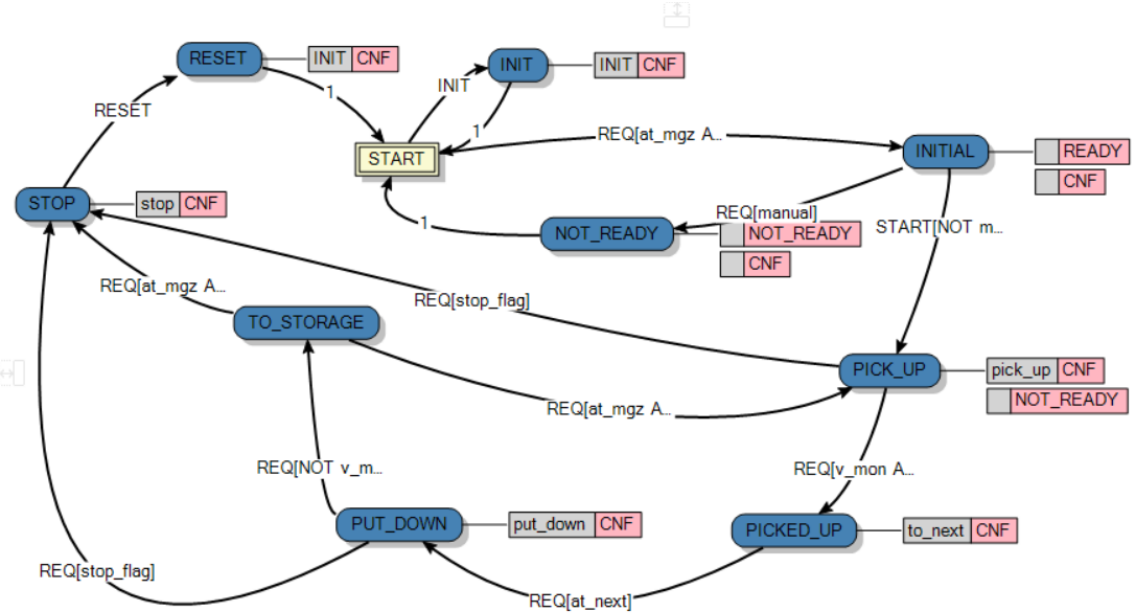


Figure 7 Arm Transport Unit State Machine

MGZ_IMC

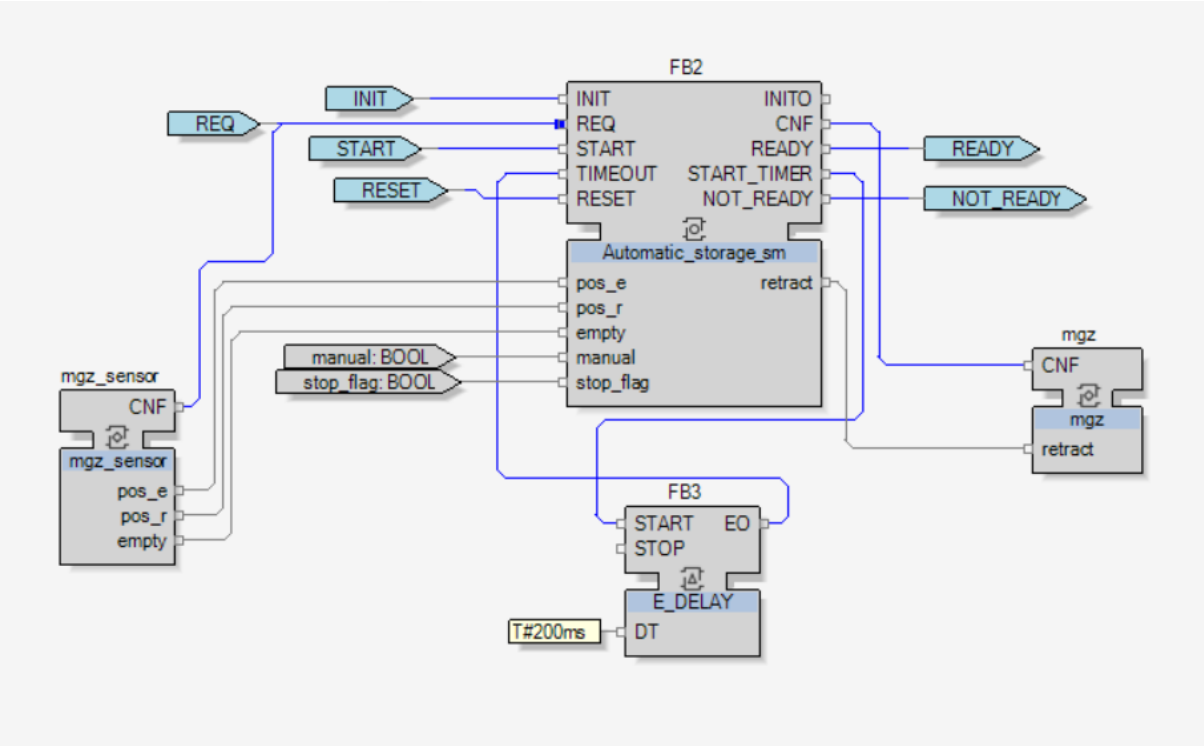


Figure 8 Stack Magazine IMC

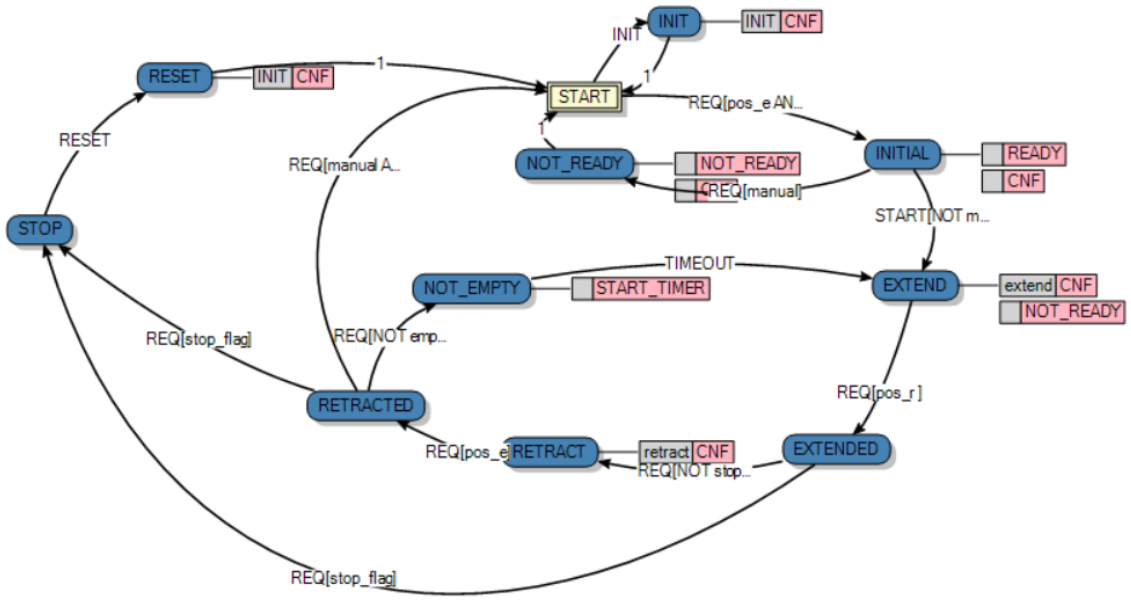


Figure 9 Stack Magazine State Machine

BUTTON_IMC

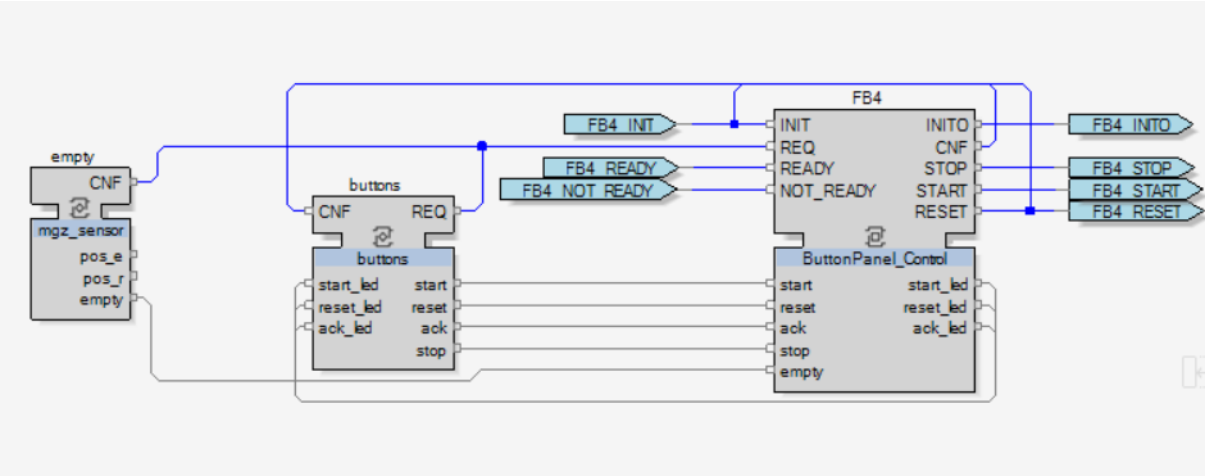


Figure 10 Button Control IMC

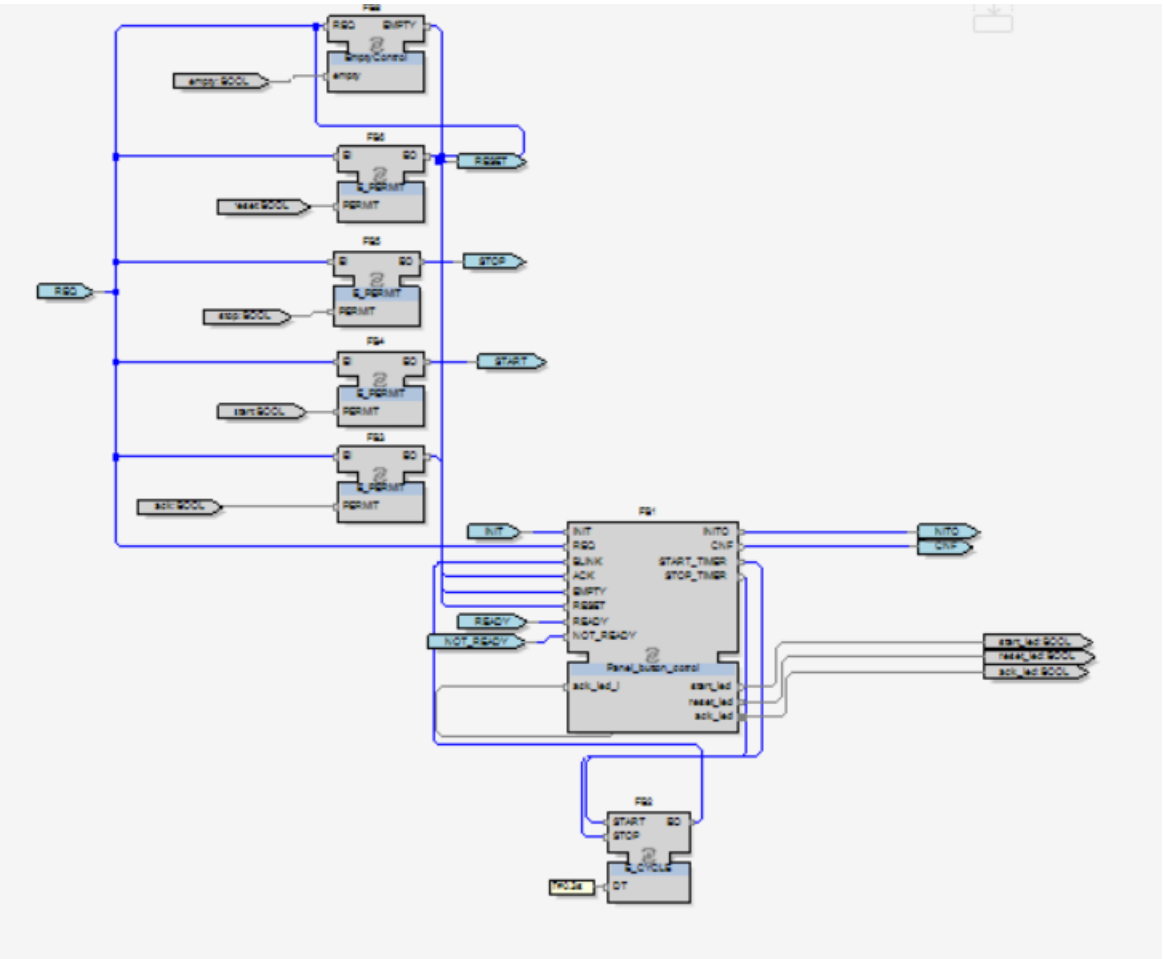


Figure 11 Button control

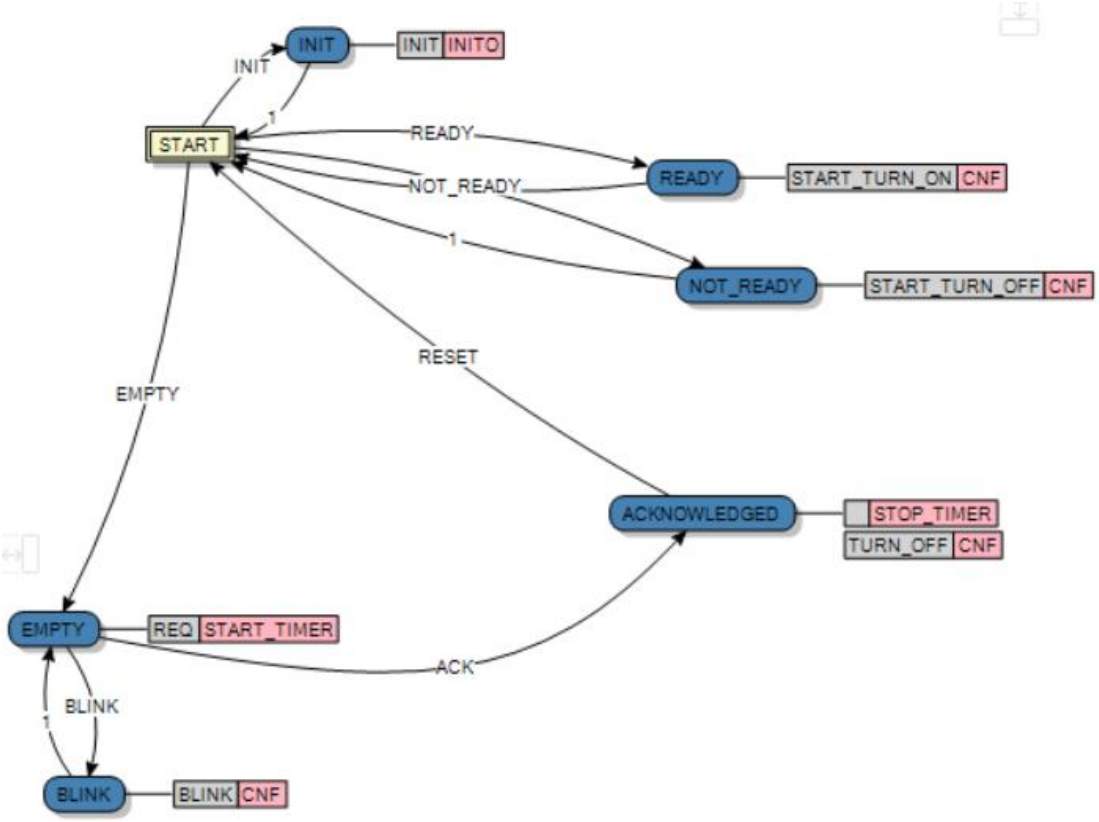


Figure 12 Button control State Machine

Supervision

The responsibilities of supervision block is to decide between manual and automated control, handle stop command and ready indication

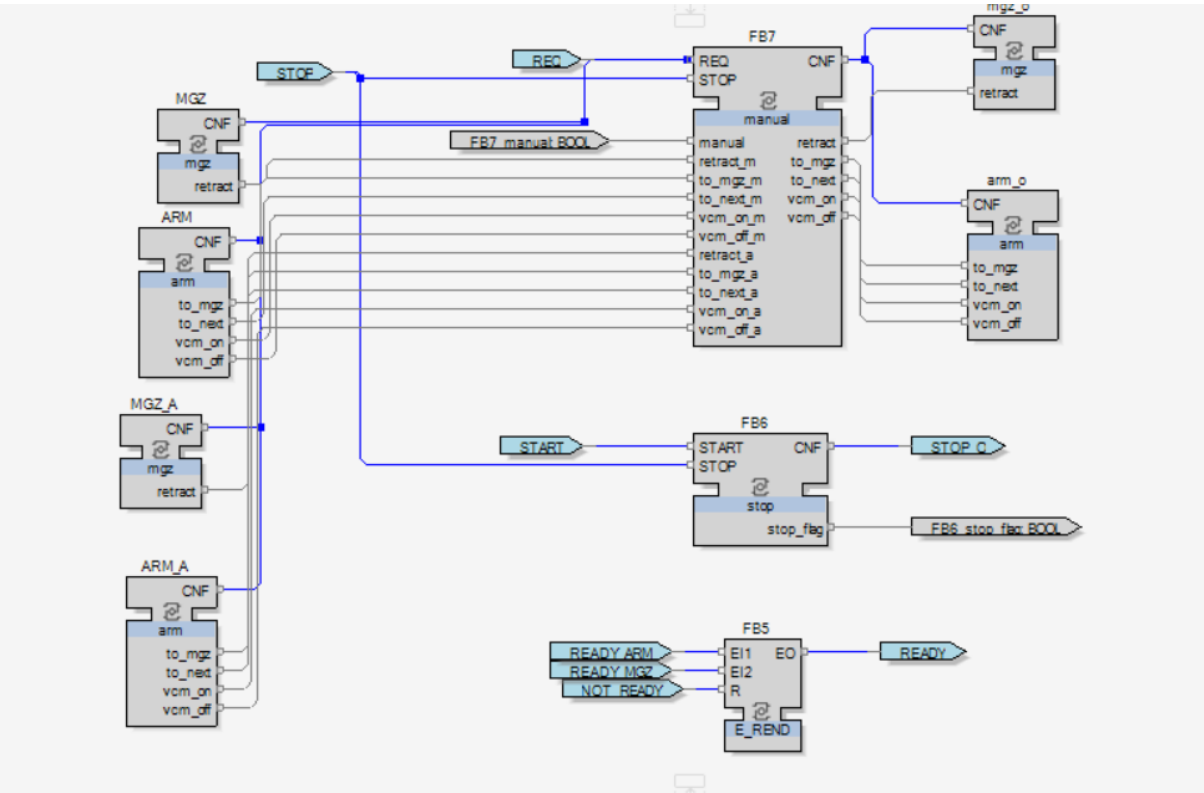


Figure 13 Supervision Architecture

6.Traceability table

Requirement	Description	Implementation
GR-01	Manual and Automatic select	Supervision block receives data both from manual and automated control. Switch Manual/Automatic sets if system will follow manual command or Automated instructions
IR-01	Initial state Stack Magazine	INIT state algorithm of Automatic_storage_sm
IR-02	Initial state Transfer Unit	INIT state algorithm of ARM_Control
IR-03	Setting initial position after the boot	INIT state of Controller blocks
AMR-01	Ready Indication	Start button led turn on when both production Units indicated Ready event (Supervision block)
AMR-02	Start button	In Automatic workflow when system is ready, Start button starts automated cycle (Supervision block)
AMR-03	Stop button	Stop button sets stop_flag data. (Supervision block)Transition to Stop state is only available from stable states(Automatic_storage_sm, ARM_Control)
AMR-04	Reset after the finish of the work	Transition to SETUP state from STOP state via Reset button (Automatic_storage_sm, ARM_Control)

AMR-05	Acknowledge led on when stack is not empty	Panel_button_control INIT state sets Ack_led to true
AMR-06	Acknowledge led blinks when stack is empty	Panel_button_control BLINK state activates changing state of led
AMR-07	Acknowledge button stops blinking, after operator pressed acknowledge button	Panel_button_control ACKNOWLEDGED state
MR-01	if sensor is empty for longer then 1.5 second, stack empty indicator should activate(ack blinking)	Panel_button_control EMPTY state activates changing state of led
TR-01	Vacuum ON during transportation of workpiece between stable points	ARM_Control checks v_mon on transitions
MMR-01	When Push WP is active, Stack Magazine cylinder is extending	Test CAT
MMR-02	When Push WP is not active, Stack Magazine cylinder is retracting	
MMR-03	When GO TO MAG is active, arm goes to magazine point	
MMR-04	When GO TO NEXT is active, arm goes to next station point	
MMR-05	When both GO TO MAG and GO TO NEXT is active, arm stops	
MMR-06	When Vacuum on is active, vacuum is active	
MMR-07	When Vacuum off is active, vacuum is not active	
MMR-08	When both VACUUM ON and VACUUM OFF is active, vacuum is not active	